

traction (RE) task. In this, the objective is to identify if a predefined semantic “*relation category*” exists (or doesn’t exist) between a pair of *subject* and *object* entities present in text. This task also presents specific challenges: the linguistics *coverage problem* due to the lack of all possible training examples of a relation class and longer text span between the entities in a sentence.

For the RE task, we use a position-aware attention model (?) that consists of a word embedding, position embedding, LSTM, attention layer, linear layer, and softmax layer. We augment the word embeddings by concatenating them with POS tag and NER category embeddings which is then fed to the LSTM to get hidden states for each word. The position embeddings for a word is derived based on the relative distance of the current word from the subject and object entities. Next, the attention layer computes the final sentence representation by focusing on both the hidden states and position embeddings. Finally, sentence representation is fed to the linear layer followed by a softmax layer for relation classification.

We perform experiments using our proposed mixed objective function on two RE datasets: *TACRED* and *SemEval-2010 Task 8* whose statistics are shown in Table 9. The POS and NER tags are computed using the Stanford CoreNLP toolkit.<sup>3</sup> Following standard convention, we report the micro-averaged  $F_1$  score on TACRED and official macro-averaged  $F_1$  score on SemEval datasets. We performed only a small number of experiments to search for the hyper-parameter values of dropout, embedding size, hidden layer size, and learning rate on the development set, all other parameters remained the same as the positional attention model of ? (?).<sup>4</sup> Our results in Table ?? show that when trained with mixed objective function, our model performs quite well, producing better results than all previously reported models despite the lack of complex task-specific hyper-parameter tuning.

## 7 Related Work

**Neural Network Methods.** Neural network models for NLP have yielded impressive results on several benchmark tasks (?; ?). To learn document features for text classification task, several methods have been proposed— ? (?) uses 1D CNNs, ? (?) uses a simple bidirectional recurrent CNN with max-pooling, ? (?) applies 2D max-pooling on top of BiLSTMs, ? (?) investigates a joint CNN-LSTM model, and JZ15a, JZ16, JZ17 apply CNNs, LSTMs, and pyramidal CNNs respectively to one-hot encoding of word sequences. An alternative approach is to first learn sentence representations followed by combining them to learn document features. To do this, ? (?) first apply a CNN or LSTM followed by a gated RNN while ? (?) learn the sentence and document features in a hierarchical manner using a self-attention mechanism.

**Semi-Supervised Learning.** SSL approaches can be broadly categorized into three types: multi-view, data aug-

| Dataset | Train  | Dev    | Test   | % Neg | $K$ | $\ell$ |
|---------|--------|--------|--------|-------|-----|--------|
| TACRED  | 68,124 | 22,631 | 15,509 | 79.5% | 42  | 36     |
| SemEval | 8,000  | –      | 2,717  | 17.4% | 19  | 19     |

Table 9: Summary statistics for RE datasets; % Neg = percentage of examples with class label of “*no relation*” between entities;  $K$  = number of classes including the *no relation* class;  $\ell$  = average length of a sentence in the dataset.

| Model                 | TACRED      |             |             | SemEval-2010 |             |             |
|-----------------------|-------------|-------------|-------------|--------------|-------------|-------------|
|                       | P           | R           | $F_1$       | P            | R           | $F_1$       |
| CNN-PE <sup>a</sup>   | <b>70.3</b> | 54.2        | 61.2        | 82.1         | 83.1        | 82.5        |
| SDP-LSTM <sup>b</sup> | 66.3        | 52.7        | 58.7        | –            | –           | 83.7        |
| PA-LSTM <sup>c</sup>  | 65.7        | 64.5        | 65.1        | –            | –           | 82.7        |
| Our Method            | 66.4        | <b>67.3</b> | <b>66.8</b> | <b>83.5</b>  | <b>84.8</b> | <b>84.1</b> |

Table 10: Model performance on TACRED and SemEval datasets; P = Precision; R = Recall; <sup>a</sup>? (?); <sup>b</sup>? (?); <sup>c</sup>? (?). For fair comparison, we only report results from models that don’t use ensembling or ranking-based approaches.

mentation, and transfer learning. First, under multi-view learning, the objective is to use multiple views of both the labeled and unlabeled data to train the model. These multiple views can be obtained either from raw text (?) or from the features (JZ15b). Second, under data augmentation, as the name implies, involves pseudo-augmenting either the features or the labels. For text classification, ? (?) performed semi-supervised training using naïve Bayes and expectation-maximization algorithms and demonstrated substantial improvements in performance. MDG16 compute embedding perturbations using adversarial and virtual adversarial approaches to improve model training. Third, under transfer learning, the approach of initializing the task-specific model weights by pretrained weights from an auxiliary task is a widely used strategy that has shown to improve the performance in tasks such as text classification (?; ?), question-answering (?), and machine translation (?; ?).

## 8 Conclusion

We show that a simple BiLSTM model using maximum likelihood training can result in a competitive performance on text classification tasks without the need for an additional pretraining step. Also, in addition to maximum likelihood, using a combination of entropy minimization, adversarial, and virtual adversarial training, we report state-of-the-art results on several text classification datasets. This mixed objective function also generalizes well to other tasks such as relation extraction where it outperforms current best models.

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<sup>3</sup><https://stanfordnlp.github.io/CoreNLP/>

<sup>4</sup><https://github.com/yuhaozhang/tacred-relation>

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